

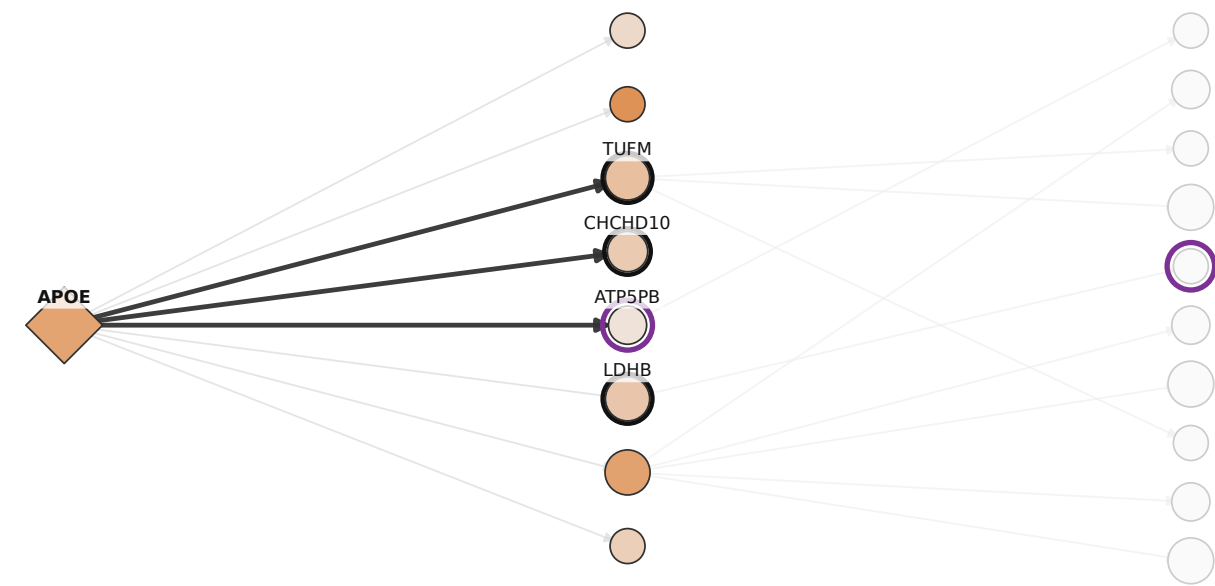
Sex-reversed mitochondrial KDA signals retain shared driver-centered network structure

Female: AD-up mitochondrial signature

Male: AD-down mitochondrial signature

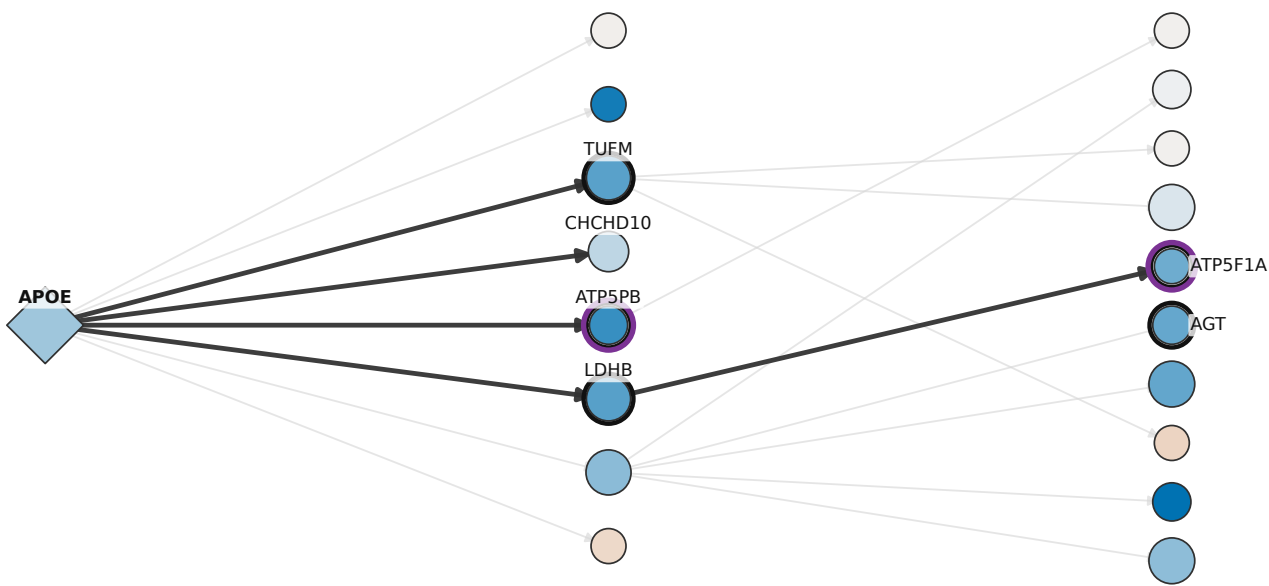
**A** APOE | Ast GRM3 | Female AD-up

APOE e2 | L=1 | n=9 | overlap=3 | FE=37.2 | q=0.0031 | shared union layout



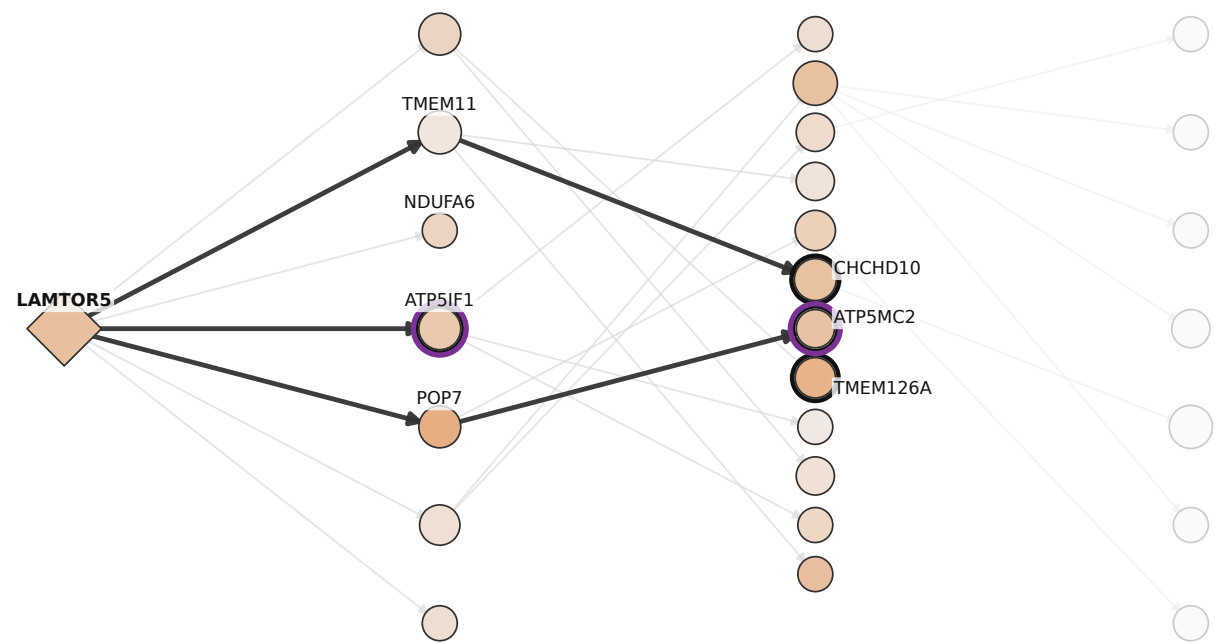
**B** APOE | Ast GRM3 | Male AD-down

APOE e2 | L=2 | n=19 | overlap=5 | FE=17.1 | q=0.00041 | shared union layout



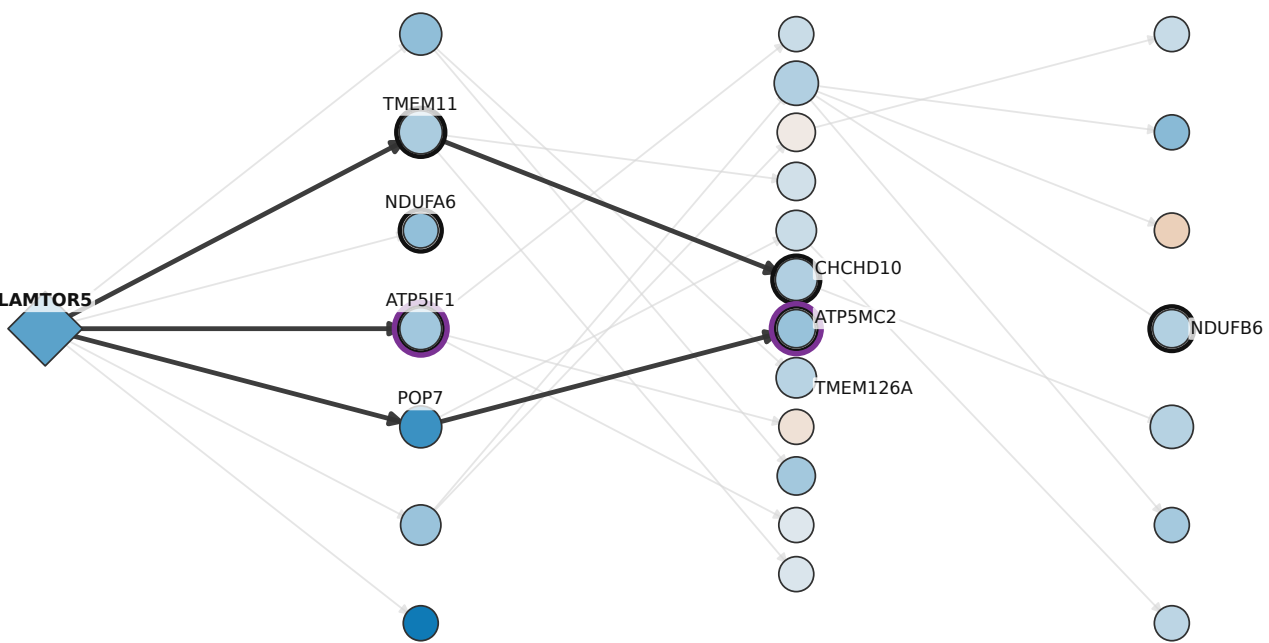
**C** LAMTOR5 | Exc L4-5 RORB IL1RAPL2 | Female AD-up

APOE e2 | L=2 | n=20 | overlap=4 | FE=19.6 | q=0.0026 | shared union layout



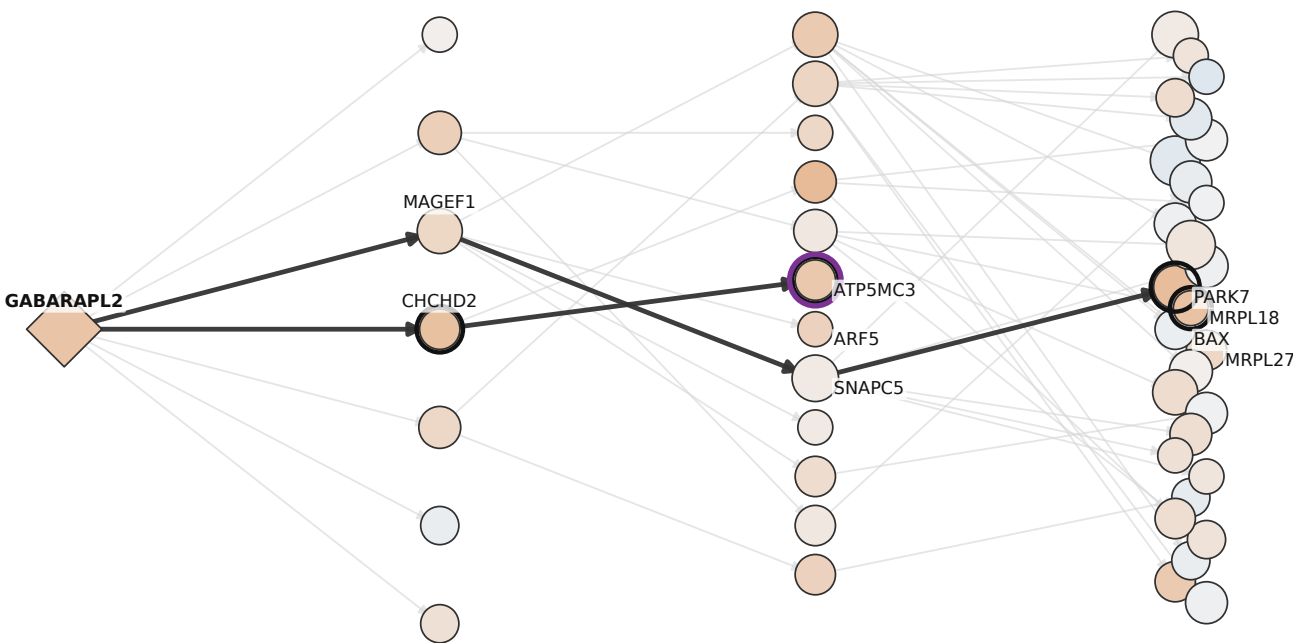
**D** LAMTOR5 | Exc L4-5 RORB IL1RAPL2 | Male AD-down

APOE e2 | L=3 | n=27 | overlap=6 | FE=19.1 | q=6.4e-05 | shared union layout



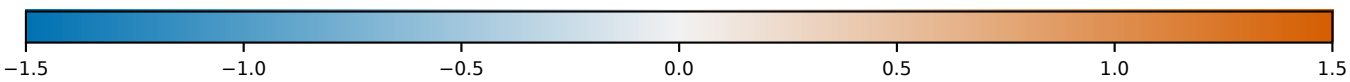
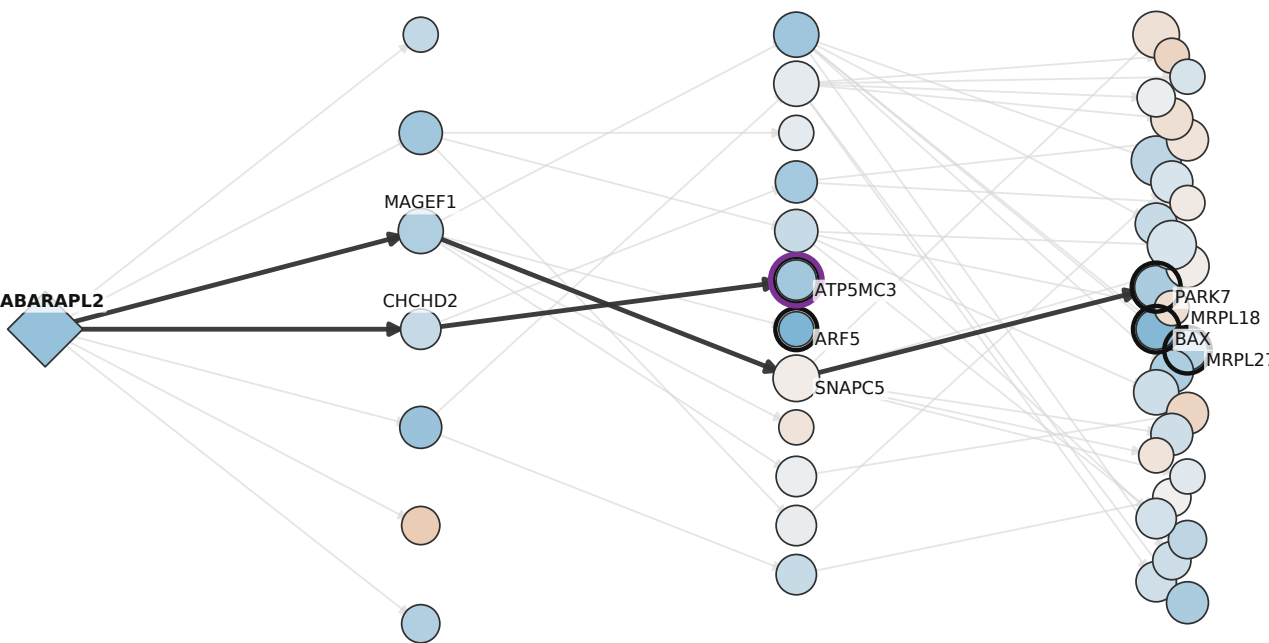
**E** GABARAPL2 | Exc L4-5 RORB IL1RAPL2 | Female AD-up

APOE e2 | L=3 | n=48 | overlap=4 | FE=8.2 | q=0.044 | shared union layout



**F** GABARAPL2 | Exc L4-5 RORB IL1RAPL2 | Male AD-down

APOE e2 | L=3 | n=48 | overlap=5 | FE=9.0 | q=0.011 | shared union layout



Columns use the same node coordinates within each row. Pale nodes/edges are absent from that condition's reconstructed KDA neighborhood; black/purple rings remain condition-specific.